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Cross-group bidirectional fusion network for 3D model classification

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ABSTRACT

3D model classification is an important task in artificial intelligence, which aims to categorize 3D models based on shapes, structures, and other features. Focusing on the issues of insufficient inter-view interaction and inefficient feature fusion in some existing multi-view 3D model classification methods, this paper proposes a 3D model classification method based on Cross-group bidirectional fusion network. Firstly, dual-branch shared Swin Transformer feature extractor is combined with feature pyramid network to achieve multi-scale hierarchical feature fusion. Secondly, the cross-group interaction module constructs an attention-based semantic dependency matrix between the two view groups, enabling context-conditioned cross-view feature association. Thirdly, bidirectional feature-wise linear modulation strategy is proposed to adaptively calibrate cross-view feature responses by dynamically generating affine transformation parameters using the global semantics of the other view group. Fourthly, a stochastic hybrid data augmentation strategy is adopted during training to enhance regularization. Experimental results show that the proposed method achieves a classification accuracy of 95.58% on the ModelNet40 dataset, verifying the effectiveness of the proposed mechanism in facilitating efficient multi-view feature fusion.

1. Introduction

3D model classification is a fundamental task in computer vision and 3D geometric analysis, and serves as a key technique for 3D scene understanding. It plays an important role in applications such as smart manufacturing, digital twins, and autonomous driving. With the rapid development of deep learning, mainstream 3D model classification methods are broadly categorized into point cloud-based, voxel-based, and multi-view-based approaches according to data representations. Among them, the multi-view approach attracts attention because it leverages mature 2D vision backbones to efficiently extract semantic cues from rendered views.

Early multi-view methods such as MVCNN (Su et al., 2015) and RotationNet (Kanezaki et al., 2018) mainly rely on view pooling for feature aggregation, which provides limited capability for explicitly modeling cross-view relationships. To enhance multi-view representations, recent research efforts improve multi-view representations in multiple directions. For example, MVTN (Hamdi et al., 2025) improves view coverage by optimizing viewpoint configurations, while Zhao et al. (2023) incorporates graph modeling and relational aggregation into more general multi-view learning to characterize dependencies between views. Meanwhile, approaches such as multi-view fusion selection and finer-grained interaction modeling continue to emerge, reflecting a shift in

research focus from simple feature aggregation to more comprehensive cross-view information utilization. Despite these advances, it remains challenging to achieve both fine-grained cross-view interaction and effective fusion of multi-scale features in a unified framework. Views rendered from different perspectives often exhibit considerable heterogeneity, and straightforward aggregation or purely global relationship modeling may lead to weak correspondence of discriminative cues across views. Moreover, many existing fusion pipelines tend to learn relatively fixed aggregation behaviors after training, and may not explicitly support context-conditioned calibration between different view contexts. As a result, complementary information is often under-exploited while redundant responses are insufficiently suppressed. To address the above issues, this paper proposes a 3D model classification method based on cross-group bidirectional fusion network (CBF-Net). This network uses a dual-branch Swin Transformer in combination with Feature Pyramid Network (FPN) to extract multi-scale hierarchical features, which effectively capture both local details and global structural information. To enable robust inter-view communication, this paper designs a cross-group multi-head attention module. This module constructs an attention-based semantic dependency matrix, which achieves context-aware feature alignment. In addition, this paper introduces the bidirectional feature-wise linear modulation (Bi-FiLM) strategy to adaptively calibrate the cross-view feature responses by dynamically generating

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affine transformation parameters from the global semantics of the opposing view groups. Finally, a stochastic hybrid data augmentation strategy is employed to alternate probabilities between Mixup and CutMix during the training process in order to enhance the diversity of the data while maintaining the consistency of the 3D structure. Main contributions of this paper are shown as follows:

- (1) A multi-scale feature extraction architecture is proposed, which integrates a shared Swin Transformer with FPN. This design effectively captures comprehensive geometric semantics from each view and boosts the expressiveness of extracted features.
- (2) A cross-group interaction module is designed to construct an attention-based semantic dependency matrix across view groups, which implements context-aware feature association across views.
- (3) The Bi-FiLM strategy is introduced for adaptive feature fusion. By dynamically generating modulation parameters from opposing views, this strategy achieves superior calibration and fusion of cross-view features.
- (4) A stochastic hybrid data augmentation strategy is developed. By Mixup and CutMix, this strategy improves model generalization.

The rest of this paper is organized as follows: Section 2 gives the related works. The framework of CBF-Net for 3D model classification is introduced in Section 3. Section 4 gives experimental results. Section 5 describes conclusions.

2. Related work

2.1. 3D model classification

3D model classification methods are mainly categorized into three types of methods based on voxels, point clouds and multi-views. The voxel-based approach discretizes the 3D model into a regular grid and spatial features are extracted using 3D convolutional neural networks. Xiong et al. (2021) proposes a registration method based on point cloud voxel grids. Plaza-Leiva et al. (2017) proposes a generalized framework for implementing and comparing supervised learning classifiers. A voxel-based neighborhood computation method is used to classify points within non-overlapping voxels of a regular grid by voxel-supported region features. Wang et al. (2019) presents a voxel-based NormalNet designed for 3D object recognition. Although voxel-based methods are effective at extracting spatial structures, they suffer from a cubic complexity trade-off. High resolutions incur prohibitive computational costs, and low resolutions lead to a loss of fine-grained geometric details.

Point cloud-based methods directly process raw point cloud data, and deep learning models are utilized to capture the geometric relationships of unordered points. PointNet (Qi et al., 2017a) and PointNet++ (Qi et al., 2017b) effectively model unordered point sets using shared multilayer perceptrons and hierarchical neighborhood learning. Recent representative works further improve performance through graph-based attention (Jing et al., 2022), dynamic convolution (Wang et al., 2022), and enhanced local-global structure modeling (Cheng et al., 2024; Wu et al., 2024), while others explore model interpretability (Feng et al., 2024). However, while point cloud-based methods directly process raw data and preserve fine geometric details, it remains challenging to model the disordered and sparse nature of point clouds effectively, and these methods are susceptible to noise.

Multi-view-based methods project 3D models into 2D images from multiple viewpoints, and feature extraction is performed using 2D convolutional networks. Various strategies are explored to enhance multi-view classification. Some approaches rely on statistical descriptors or ensemble-based fusion. For instance, Gao et al. (2024) combines Shannon-entropy features with a voting mechanism, while Shi et al. (2022) employs ensemble learning with least-squares regression. In specific application scenarios, Yu et al. (2024) applies multi-view learning to mitigate occlusion-related challenges. More recent studies make notable progress in view optimization. Hamdi et al. (2025) introduces the

MVTN, which leverages differentiable rendering to learn informative viewpoints, demonstrating advantages over fixed-view settings. Compared to voxel and point cloud paradigms, multi-view methods strike a balance by offering efficient computational performance and scalability. While recent advances improve viewpoint selection and representation learning, it remains challenging to achieve fine-grained cross-view interaction and hierarchical multi-scale fusion under heterogeneous views. These challenges motivate our discussions on multi-view feature fusion and cross-group interactions in the next sections.

2.2. Multiview feature fusion

The core challenge of multi-view feature fusion lies in balancing complementarity and redundancy among views while achieving effective information aggregation. Early research, such as Zhao et al. (2023), proposes a Reinforced Tensor Graph Neural Network to improve graph-based multi-view representation. More recent studies explore dynamic association, reliability constraints, and automated architecture design. For instance, Liu et al. (2025) emphasizes multi-level information fusion in complex graph structures. To address view quality variations, Evolutionary Multi-View Classification (Liang et al., 2021) eliminates individual fitness bias to improve robustness, while Trusted Multi-View Classification (Liang et al., 2025b) incorporates expert-knowledge constraints. Furthermore, view-association-guided classification (Liang et al., 2025a) explores dynamic mechanisms to associate relevant views, and LogicNAS (Guo et al., 2025) introduces neural architecture search to optimize fusion structures. In the 3D domain, Gao et al. (2025) utilizes deep residual shrinkage networks to enhance noise robustness during fusion.

Despite these recent advances, many existing frameworks still face limitations in interaction granularity and hierarchical processing. Moreover, many fusion pipelines provide limited inter-view interaction depth, since they rely on coarse-grained aggregation or shallow message passing at a single scale. As a result, cross-view feature alignment is often weak, and discriminative cues are not consistently matched across views during fusion. Local geometric details and global structural semantics often require different fusion strategies. It is common for cross-view interactions to remain unidirectional or static in many pipelines, which limits context-conditioned feature calibration across views. To address these limitations, this paper proposes a multi-scale pyramid fusion framework integrated with Bi-FiLM. By establishing local-global associations at four distinct scales and enabling bidirectional modulation, our method promotes effective cross-view feature association and fusion across scales.

2.3. Cross-group feature interactions and modulation

To improve cross-view interaction modeling, prior studies introduce attention mechanisms and conditional modulation to facilitate information exchange across branches. FiLM (Perez et al., 2018) is a general-purpose conditioning method that modulates intermediate feature maps via predicted affine parameters, and is extended to graph neural networks (Brockschmidt, 2020) and dense prediction tasks (Lemay et al., 2021). In parallel, self-attention is widely adopted to capture dependencies among feature tokens.

However, it remains challenging to perform multi-view 3D classification under significant inter-view heterogeneity, because effective fusion should account for both fine-grained interactions and hierarchical representations. In many pipelines, generic attention or modulation designs may exhibit two common limitations. Firstly, interactions are often dominated by global representations or unidirectional conditioning, so inter-view communication often lacks fine-grained interaction depth beyond global summaries. Secondly, interaction and fusion are frequently decoupled from multi-scale hierarchies. Meanwhile, without explicit mechanisms for feature alignment, complementary cues across views are easily misaligned and weakened during aggregation.

To address these issues, we integrate cross-group interaction with Bi-FiLM. Cross-group attention establishes cross-view dependency modeling between view groups in the feature space. Meanwhile, modulation parameters are generated from the global semantics of the opposing view group, enabling bidirectional and context-conditioned calibration that helps suppress redundant responses and reinforces complementary information.

2.4. Data augmentation

Data augmentation is a pivotal technique for enhancing model generalization. Methods such as Mixup (Zhang et al., 2017) and CutMix (Yun et al., 2019) are effective in general vision tasks, which motivates further improvements and extensions (Fisher et al., 2024; Sun et al., 2020). However, it is challenging to directly apply these 2D image augmentations to multi-view 3D recognition. Since a single object is characterized by multiple views, cross-view inconsistencies easily arise when aggressive mixing or region replacement is applied independently to different views. These inconsistencies can disrupt the stable aggregation of discriminative cues. Overly conservative augmentation may fail to provide sufficient regularization to suppress overfitting. For instance, while Attentive CutMix (Walawalkar et al., 2020) focuses on salient local regions, it does not explicitly enforce cross-view consistency in multi-view settings. Consequently, it is beneficial to adopt a stochastic hybrid augmentation strategy that switches between Mixup and CutMix probabilistically during training. This approach aims to bolster sample diversity and regularization while mitigating excessive cross-view inconsistencies, thereby improving model generalization.

3. CBF-net for 3D model classification

This paper proposes a new 3D model classification framework based on CBF-Net as shown in Fig. 1.

3.1. Multi-view extraction and grouping of 3D models

This paper uses airplane model in ModelNet40 dataset, and takes $n=12$ as an example. The projection views of 3D model from different angles are shown in Fig. 2.

Multi-view extraction and grouping of 3D model M is achieved through a standardized process: a unified visual environment is constructed with the Phong light model. M is scaled to a specified range and 12 virtual cameras are deployed uniformly along a circular track. Each camera view is spaced 30° apart. Based on the above lighting and camera parameters, a 2D view set $V_M = \{v_1, v_2, \dots, v_n\}$ is generated. Here, $1 \leq n \leq 12$, v_i denotes the image acquired from the i -th viewpoint. The final 2D view set of V_M is divided into two groups: Group1 = $\{v_1, v_2, \dots, v_6\}$ and Group2 = $\{v_7, v_8, \dots, v_{12}\}$.

3.2. Multi-scale feature extraction based on swin transformer

The quality of feature extraction has significant impact on 3D model classification. In this paper, we adopt shared Swin Transformer as the feature extraction backbone network to realize hierarchical feature extraction and multi-scale fusion for multi-view groups. The images of the two view groups Group1 and Group2 are input into shared Swin Transformer network. This network divides the images into non-overlapping patches and embeds the features by PatchEmbed, and generates 4 stages of hierarchical features ($F_1 \sim F_4$) corresponding to different scales by multiple BasicLayers, as shown in formula (2).

$$x_{patch} = \text{Flatten}(\text{Conv}(x)) \quad (1)$$

$$F_i = \text{SwinStage}_i(x_{patch}), \quad i = 1, 2, 3, 4 \quad (2)$$

Here, $F_1 \sim F_4$ correspond to the features labeled as $S_2 \sim S_5$ in Fig. 3, representing the outputs of PatchEmbed and the first three BasicLayers respectively.

To integrate the multi-scale features of different stages, we apply a top-down FPN structure to the 4 stage features $F_1 \sim F_4$ of each view group. Firstly, channel alignment is performed on the features of each stage using 1×1 convolution and preserving their spatial location information to generate transversally connected features. Secondly, starting from Stage 3, the output P_{i+1} from the previous stage is scaled up to the spatial size of the current stage by nearest-neighbor up-sampling, and then summed with the transversally connected features of the current stage in order to fuse the high-level semantics with the low-level details. Thirdly, the set of output features at 4 scales $P = \{P_1, P_2, P_3, P_4\}$ as shown in formula (3).

$$\begin{cases} P_4 = \phi_4(F_4), \\ P_i = \phi_i(F_i) + \text{Up}(P_{i+1}) \end{cases} \quad (3)$$

Here, ϕ_i represents 1×1 convolution operation, $\text{Up}(\cdot)$ denotes nearest neighbor up-sampling.

The top-down fusion process in formula (3) is visualized in the core structure of Swin Transformer-FPN shown in Fig. 3.

After obtaining the multi-scale features P_i of each view group, FPN features P_i within group g ($g = 1, 2$) are weighted and summed by the learnable weights. α_v^g is obtained by normalizing the weight parameter w_v^g to achieve view aggregation, as shown in formula (4).

$$\alpha_v^g = \frac{\exp(w_v^g)}{\sum_{u=1}^N \exp(w_u^g)} \quad (4)$$

Here, g denotes two view groups. w_v^g represents the learnable weight for the v -th view in group g , and N is the total number of views for each group.

The view aggregation process for group g on the i -th scale is shown in formula (5).

$$A_i^g = \sum_{v=1}^N \alpha_v^g P_i^{(v,g)} \quad (5)$$

Here, $P_i^{(v,g)}$ represents the features of the v -th view of the i -th scale in group g .

3.3. Cross-group attention interaction and feature modulation

The proposed cross-group bidirectional fusion mechanism differs from a simple combination of cross-attention and FiLM. Standard cross-attention mainly computes dependencies between two feature sequences, while standard FiLM usually performs one-way conditional affine modulation. General multi-branch fusion methods often aggregate branch features by concatenation, addition, pooling, or attention weighting, without explicitly coupling cross-view dependency modeling and feature recalibration. In contrast, CBF-Net organizes rendered views into two complementary groups and performs group-level cross-view interaction. At each scale, cross-group attention first models semantic dependencies between the two view groups, and Bi-FiLM then generates affine parameters from the global contexts of the view groups to recalibrate the interacted features in a bidirectional manner. This design is particularly suitable for multi-view 3D classification, where different views contain complementary but also redundant shape cues. Therefore, the proposed mechanism enables cross-view feature association and context-conditioned calibration before final aggregation.

In multi-view 3D model classification, a key challenge is to effectively model geometric dependencies between views and achieve balanced feature fusion. Direct feature concatenation or pooling may be insufficient to capture complex cross-view correlations, while attention alone does not explicitly perform feature-wise response calibration. To address this issue, the proposed module couples cross-group attention with Bi-FiLM, thereby performing semantic dependency modeling and conditional modulation jointly before final aggregation.

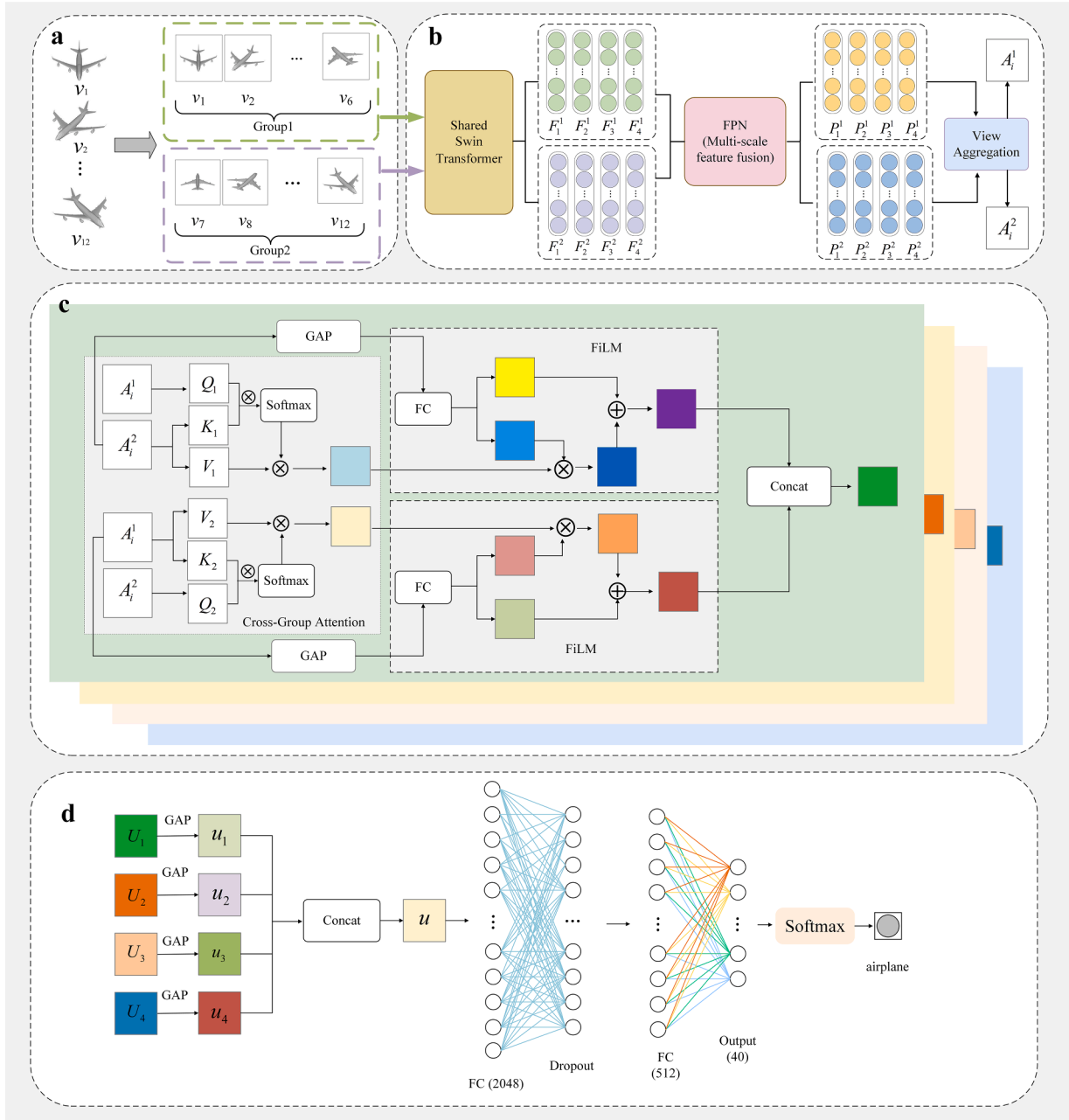


Fig. 1. The framework of 3D model classification.

The aggregated features A_i^1 and A_i^2 of Group1 and Group2 at the same scale i are respectively used as Query, Key, and Value to realize feature fusion through multi-head cross-attention. The specific steps are as follows.

(1) The aggregated features A_i^1 and A_i^2 are spread as sequences, as shown in formula (6).

$$\begin{aligned} Q_1 &= \text{Vec}(A_i^1), & K_1 &= \text{Vec}(A_i^2), & V_1 &= \text{Vec}(A_i^2) \\ Q_2 &= \text{Vec}(A_i^2), & K_2 &= \text{Vec}(A_i^1), & V_2 &= \text{Vec}(A_i^1) \end{aligned} \quad (6)$$

Here, $\text{Vec}(\cdot)$ operation spreads the spatial dimensions into sequence lengths to facilitate attention computation.

(2) Interaction features are computed through a multi-attention mechanism, as shown in formula (7).

$$\text{CrossAttn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (7)$$

Here, d represents the feature dimension. This operation calculates the similarity between Query and Key by scaling the dot product attention. Then the Value is weighted and summed to get the interaction features.

(3) Restore the attentional output to the spatial dimension, as shown in formula (8).

$$C_i = \text{Reshape}(\text{CrossAttn}(Q, K, V)) \quad (8)$$

(4) Global average pooling is performed for each set of aggregated features to extract the global condition vector, as shown in formula (9).

$$c_i^g = \text{GAP}(A_i^g) \quad (9)$$

Here, c_i^g denotes the global condition vector of the g -th group on the i -th scale, which is extracted from A_i^g by global average pooling.

(5) The affine transformation coefficients γ and β are generated using the fully connected (FC) layer, as shown in formula (10).

$$[\gamma_i^g, \beta_i^g] = \text{FC}_i(c_i^g) \quad (10)$$

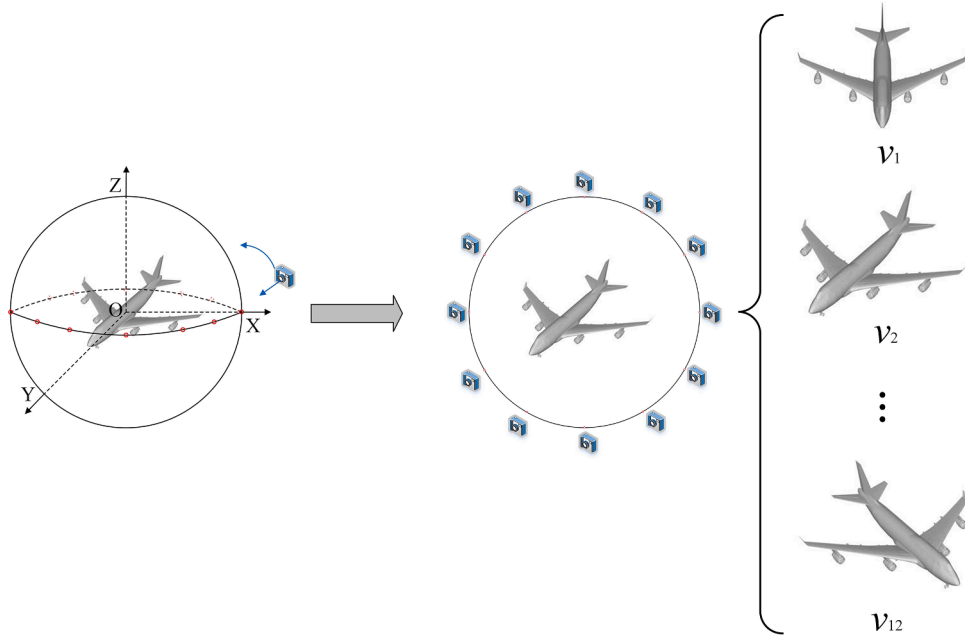


Fig. 2. Projected views of the airplane model.

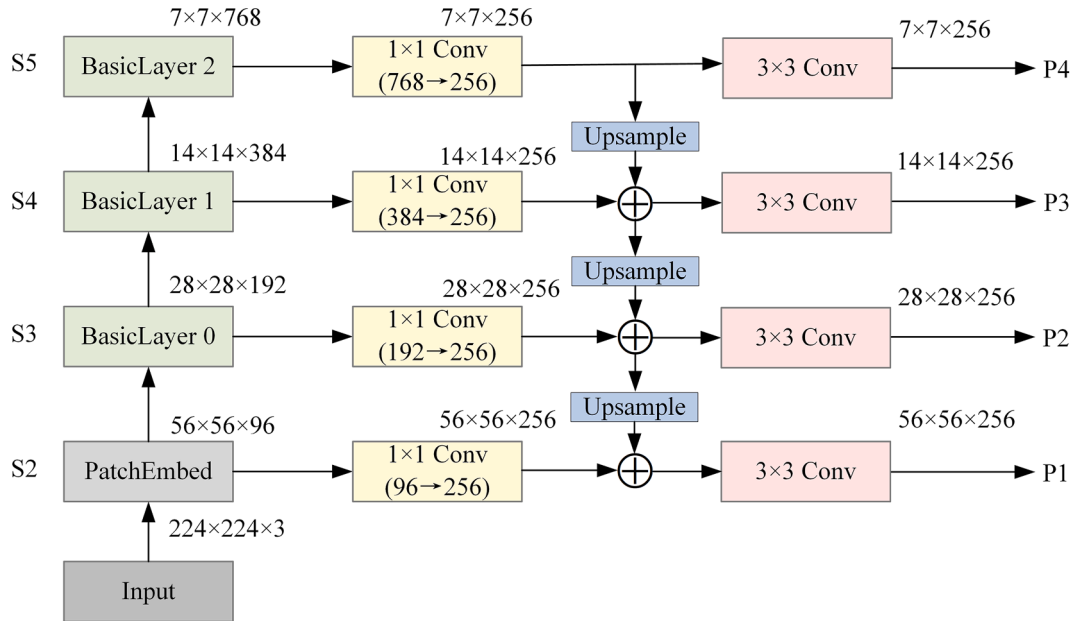


Fig. 3. The core structure of Swin Transformer-FPN.

Here, FC_i represents FC layer corresponding to the i -th scale.

(6) The interaction feature C_i is modulated using the affine coefficients of the other group, as shown in formula (11).

$$M_i^1 = \gamma_i^2 \odot C_i + \beta_i^2, \quad M_i^2 = \gamma_i^1 \odot C_i + \beta_i^1 \quad (11)$$

Here, $\odot$ represents element-by-element multiplication.

(7) The bidirectional FiLM-modulated features of the i -th scale are concatenated along the channel dimension to form the single-scale aggregated feature U_i , as shown in formula (12).

$$U_i = [M_i^1; M_i^2] \quad (12)$$

This process realizes inter-group feature interaction through cross-group attention. It enhances feature complementarity by combining with Bi-FiLM modulation, which effectively improves the discriminative

ability of features and provides more discriminative feature representations for 3D model classification.

3.4. Multi-scale aggregation and classification

Each single-scale aggregated feature U_i is converted into a 1D global vector u_i by global average pooling and flattening. The multi-scale features U_1, U_2, U_3 , and U_4 are fused by concatenating their corresponding global vectors u_1, u_2, u_3, u_4 along the channel dimension, forming the unified multi-scale representation u , as shown in formula (13).

$$u = [u_1, u_2, u_3, u_4] \quad (13)$$

The unified vector u is directly fed into a FC layer to obtain $FC_M = \{f_j \mid 0 \leq j \leq c-1\}$. Then input FC_M into softmax function to compute

Table 1

Comparison with representative 3D model classification methods on ModelNet40.

Methods	Accuracy (%)
MVCNN (Su et al., 2015)	90.10
MVTN (Hamdi et al., 2025)	93.80
MV-DRSN (Gao et al., 2025)	91.12
PointCMT (Yan et al., 2022)	94.40
DGACN (Huang et al., 2024)	94.10
MVT (Chen et al., 2021)	94.40
EFSCNN (Zhou et al., 2023)	92.30
CCL-3D (Long et al., 2025)	93.40
The proposed network	95.58

the probability $P(s_j | M)$ that M belongs to the category s_j , as shown in formula (14).

$$P(s_j | M) = \frac{e^{f_j}}{\sum_{i=0}^{c-1} e^{f_i}} \quad (14)$$

Category s of M is determined as shown in formula (15).

$$s = \arg \max_{j=0,1,\dots,c-1} P(s_j | M) \quad (15)$$

4. Experiments

Experiments are performed on the publicly available dataset ModelNet40 to evaluate the performance of the methods in this paper on the 3D model classification task. ModelNet40 has 40 categories and 12,311 3D models. There are 9843 models in training set. Test set contains 2468 models.

4.1. Performance comparison with representative methods

This section compares the proposed CBF-Net with several representative 3D model classification methods on the ModelNet40 dataset. The experimental results are shown in Table 1.

As shown in Table 1, CBF-Net achieves 95.58% accuracy on ModelNet40 under the rendering protocol adopted in this paper. Compared with the representative methods listed in the table, the proposed method shows competitive performance. It should also be noted that recent view-based methods such as View-GCN (Wei et al., 2020) and ViewFormer (Sun et al., 2023) report higher accuracies under their respective experimental settings. Therefore, this paper does not claim that CBF-Net outperforms all recent state-of-the-art methods. Since published results in multi-view 3D recognition may be affected by rendering settings, view numbers, input resolutions, pretraining strategies, and evaluation protocols, Table 1 is presented only as a reference comparison with representative published methods.

4.2. The selection of feature extraction networks

The group of experiments are performed to investigate the influence of different Swin Transformer networks on the extraction of view features. Different Swin Transformer networks can be obtained by specifying the dimension of the embedding and the number of blocks contained in stage3. Training set of 2D views is used to optimize the proposed network, and test set is adopted to verify the classification performance of the different Swin Transformer networks. The classification results of different Swin Transformer networks are shown in Table 2.

From Table 2, it can be seen that the performance of Swin_small is the best and its accuracy is 93.16%. Swin_base achieves the worst and its accuracy is 91.93%. The accuracy of Swin_small is slightly higher than Swin_tiny. However, Swin_tiny has lower MACs and number of parameters, faster training speed, and less resource consumption. In order to take into account the classification effect and efficiency, Swin_tiny is chosen as Swin Transformer in the later experiments of this paper.

In order to observe the effect of different Swin Transformer networks on view feature extraction more intuitively. The raw data of Embedding Dimension, Depth (Stage 3), MACs, and Params in Table 2 are normalized to [0, 1] using the Min-Max Normalization method. The radar plot is shown in Fig. 4.

From Fig. 4, it can be seen that the MACs value of swin_tiny is significantly lower than the other models, which indicates that its memory access cost is lower and its computational efficiency is better. swin_tiny has the least number of parameters, which means that it consumes fewer resources and the training speed is faster.

The network for extracting view features has a significant impact on 3D model classification performance. In the group of experiments, VGG-16, ResNet50, DenseNet121, and Swin Transformer are selected as feature extraction networks. All networks are not initialized with pre-trained weights, and the rest of the parameters remain the same. Training set is used to optimize VGG-16, ResNet50, DenseNet121, and Swin Transformer. Test set is adopted to testify the optimized VGG-16, ResNet50, DenseNet121, and Swin Transformer. Accuracies are shown in Table 3.

From Table 3, it can be seen that DenseNet121 achieves better than VGG-16 and ResNet50 on classification accuracy. Classification accuracies of VGG-16, ResNet50, and DenseNet121 are respectively 90.06%, 89.04%, and 90.26%. It indicates that DenseNet121 can extract more effective view features than VGG-16 and ResNet50. Classification accuracy of Swin Transformer is higher than that of VGG-16, ResNet50, and DenseNet121. Classification accuracy of Swin Transformer reaches 90.92%, which is 0.66% higher than that of DenseNet121. This is because Swin Transformer adopts a hierarchical architecture and self-attention mechanism. When processing multi-view features, the self-attention mechanism can break through the limitation of the local receptive field of traditional convolutional neural networks, thus extracting more discriminative 3D model classification features. Therefore, Swin Transformer achieves better than DenseNet121.

4.3. Ablation analysis of view input and grouping strategies

4.3.1. Impact of view number and selection strategy

The group of experiments includes Experiments 1–4. This group of experiments uses Swin Transformer as the base feature extractor with FiLM module to conduct experiments based on different number of views and input strategies. In Experiment 1, 1 view from each of the two branches is randomly selected as an input feature. In Experiment 2, 6 views are randomly selected as input features for each of the two branches. In Experiment 3, one branch fixes the selection of the first 6 views, and the other branch selects the last 6 views as input features. In Experiment 4, 12 views are selected as input features for both branches. Training set is used to optimize the proposed network and test set is adopted to testify it. Accuracies are shown in Table 4.

From Table 4, it can be seen that Experiment 3 achieves the best with accuracy of 93.76%. The accuracy of Experiment 2 is 93.35%, which is slightly lower than that Experiment 3. The reason is that there is uncertainty in the random selection, and the strategy of fixing the selection of the first 6 views and the last 6 views makes the view selection more regular and complementary. Experiments 1 and 4 performed poorly with accuracy of 92.42% and 92.46% respectively. The reason is that 1 view random selection is adopted in Experiment 1, making it difficult to capture complete features. In Experiment 4, 12 views are selected for both branches, which increases redundant information and introduces noise interference. Therefore, the view input strategy employed in the subsequent experiments of this paper is that the first 6 views are fixedly selected for one branch, while the last 6 views are selected for the other branch.

4.3.2. Sensitivity analysis of view grouping strategies

To assess the sensitivity of CBF-Net to view grouping strategies, the group of experiments includes seven comparative settings (S1-S7) within

Table 2
Accuracies of different Swin Transformer Networks.

Network	Embedding Dimension	Depth (Stage3)	Number of Heads	MACs (G)	Params (M)	Accuracy (%)
Swin_tiny	96	6	(3,6,12,24)	4.27	28.27	92.97
Swin_small	96	18	(3,6,12,24)	8.34	49.56	93.16
Swin_base	128	18	(4,8,16,32)	14.81	87.70	91.93
Swin_large	192	18	(6,12,24,48)	33.32	228.47	92.55

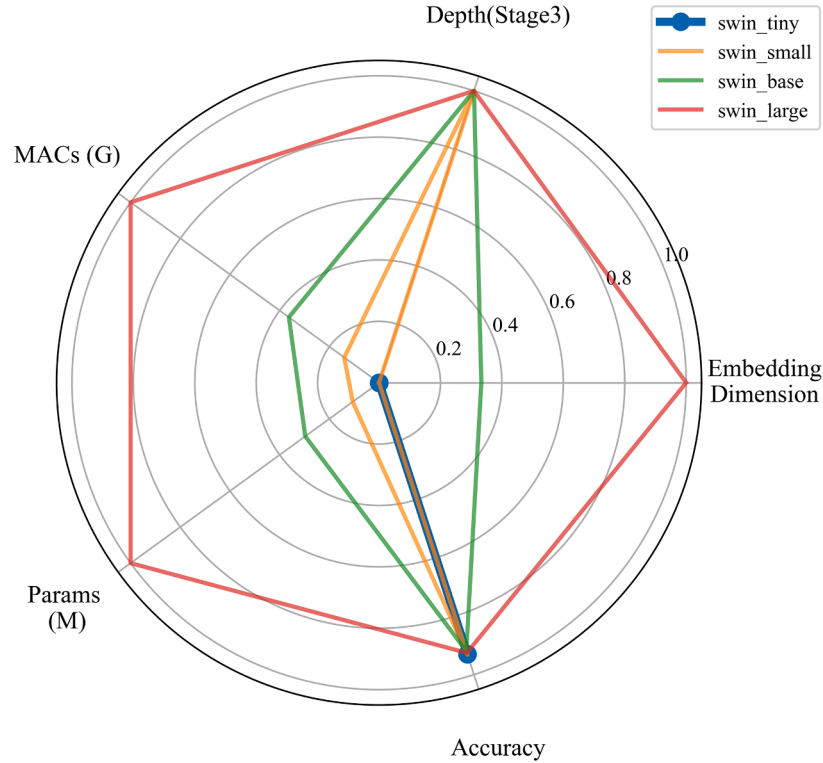


Fig. 4. Structural parameters and classification performance of different Swin Transformer Networks.

Table 3
Accuracies of different feature extraction networks.

Network	Accuracy (%)
VGG-16	90.06
ResNet50	89.04
DenseNet121	90.26
Swin Transformer	90.92

Table 4
Accuracies of experiments with different view numbers and input strategies.

Experiment	Accuracy (%)
Exp 1	92.42
Exp 2	93.35
Exp 3	93.76
Exp 4	92.46

the full network framework. In S1, the 12 views are evenly divided into two groups while preserving their original order, which corresponds to the default setting of CBF-Net. S2 uses a 4/8 split and S3 uses an 8/4 split to investigate the effect of imbalanced group sizes. In S4, a circular shift of three views is applied to the input sequence before fixed grouping to evaluate sensitivity to the starting viewpoint. S5 adopts an alternating grouping scheme based on view index parity, while S6 uses random grouping to disrupt the view order. As a baseline, S7 employs a

single-branch 12-view fusion model without the cross-group interaction module. The classification accuracies are shown in Fig. 5.

As shown in Fig. 5, S1 achieves the highest accuracy of 95.58%. It outperforms the single-branch baseline S7 by 0.89 percentage points, which demonstrates that introducing view grouping together with cross-group interaction benefits the discriminative capability of multi-view fusion in CBF-Net. In contrast, the random grouping strategy S6 achieves 94.25%, which is the lowest among all settings. This indicates that without stable view organization constraints, cross-group interactions may find it harder to learn consistent correspondences, thereby diminishing fusion benefits. Moreover, strategies with structured grouping generally outperform random grouping, demonstrating the importance of reasonable view organization. For imbalanced grouping, S2 and S3 obtain 94.37% and 94.81% respectively, which are both lower than S1. This indicates that the balanced group sizes better support cross-group interaction and modulation. S4 achieves 95.26%, with a small gap compared to S1, which indicates that the model is relatively insensitive to changes in the starting view. S5 achieves 95.22%, maintaining competitive performance even without local continuity, which indicates that CBF-Net can adapt to alternative grouping topologies. Overall, S1 is adopted as the default grouping strategy.

4.4. Ablation experiments with cross-group interactive components

The group of experiments includes Experiments 5–7, in order to explore the effect of cross-group attention on enhancing feature discriminability. In Experiment 5, cross-group feature fusion is achieved using single FiLM and multi-head cross-attention. In Experiment 6, cross-

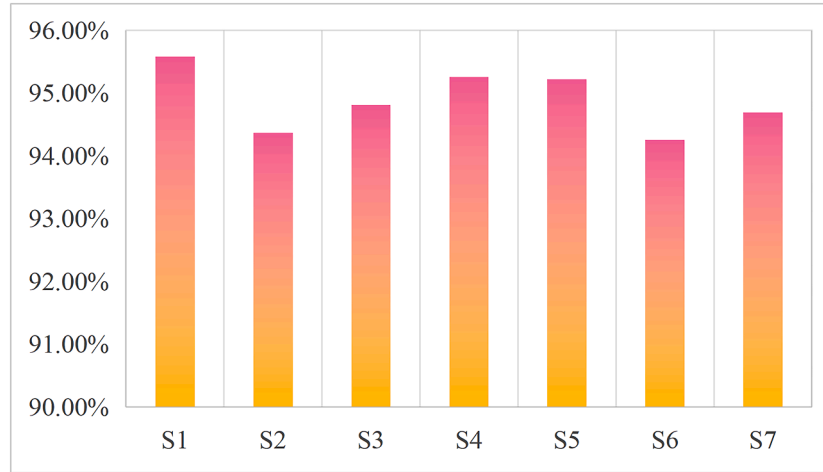


Fig. 5. Classification accuracy under different view grouping strategies.

Table 5

Accuracies of different cross-group feature fusion methods.

Experiment	Accuracy (%)
Exp 3	93.76
Exp 5	93.92
Exp 6	94.12
Exp 7	93.88

Table 6

Accuracies of different multi-scale feature fusion strategies.

Methods	Accuracy (%)
Simplified-FPN	94.89
U-Net Skip Connections	94.61
The proposed network	95.58

group feature fusion is realized by combining Bi-FiLM with multi-head cross-attention. In Experiment 7, cross-group feature fusion uses Bi-FiLM and single-head attention mechanism. Accuracies are shown in Table 5.

From Table 5, it can be seen that Experiment 5 achieves better than Experiment 3 on accuracy. Accuracies of Experiment 3 and Experiment 5 are respectively 93.76% and 93.92%. It indicates that Experiment 5 can extract more effective features than Experiment 3. Accuracies of Experiment 5, Experiment 6 and Experiment 7 are respectively 93.92%, 94.12% and 93.88%. This shows that Bi-FiLM combined with multiple cross-attention has a better ability to extract complementary information from between views. The reason is that multiple cross-attention is better at capturing multi-perspective relationships between groups, and Bi-FiLM can more fully explore information about intergroup complementarities.

4.5. Ablation analysis of multi-scale feature fusion

4.5.1. Impact of fusion strategies

The group of experiments explores the influence of different multi-scale feature fusion strategies on the classification performance of 3D models. This set of experiments is conducted using simplified-FPN, U-Net skip connections and the FPN proposed in this paper respectively. Training set is used to optimize 3 networks, and test set is used to verify accuracies of these 3 networks. Accuracies are shown in Table 6.

From Table 6, it can be seen that Simplified-FPN achieves better than U-Net skip connections on 3D model classification task. Accuracies of Simplified-FPN and U-Net skip connections are respectively 94.89% and 94.61%. The reason is that Simplified-FPN independently downscales

Table 7

Accuracies of different scale-level multi-scale fusion weighting schemes.

Methods	Accuracy (%)
Learnable Scale Weighting (Softmax)	94.85
Equal Scale Weighting	95.58

the features at each stage by combining 1×1 convolution through the lateral connection structure. It avoids checkerboard artifacts introduced by decoding transposed convolution in U-Net skip connection and the deficiency of fixed concatenation on the spatial focusing ability of small targets and realizes more efficient lightweight fusion of multi-scale features. The proposed network achieves the highest accuracy. The reason is that compared with Simplified-FPN and U-Net skip connections, the FPN proposed in this paper employs bidirectional feature fusion through lateral connections and up-sampling. This design strengthens cross-level feature interactions and enables more targeted optimization of multi-scale feature extraction.

Confusion matrix is constructed using the data in Table 6 to analyze the misclassification of each category by the proposed network, as shown in Fig. 6.

From Fig. 6, it can be seen that accuracies of airplane, car, cone, curtain, door, guitar, keyboard, laptop, mantel, sink, stairs, toilet, xbox categories achieve 100%. However, 3D models of the flower_pot, plant, and vase categories are very similar in structure, and thus misclassification often occurs in experiments. As shown in Fig. 7, the boundaries between these models are unclear. For example, most of the models in the flowerpot category contain plants, while it remains ambiguous whether the models in the plant category contain flowerpots. The accuracy of the proposed network is affected by the mutual misclassifications between different categories.

4.5.2. Impact of multi-scale fusion weighting

The group of experiments investigates the impact of different scale-level weighting schemes in multi-scale feature fusion on the classification performance. This paper compares the equal scale weighting strategy with a learnable scale weighting strategy. In the learnable scale weighting strategy, a learnable scalar weight is introduced for each scale output by the FPN, which is normalized via the Softmax function and optimized end-to-end during training. Except for the scale-level weighting scheme, all other experimental settings remain consistent with the proposed model. The experimental results are shown in Table 7.

From Table 7, the learnable scale weighting scheme achieves 94.85%, which is lower than the 95.58% obtained by equal scale weighting. This suggests that, under our current multi-scale design and

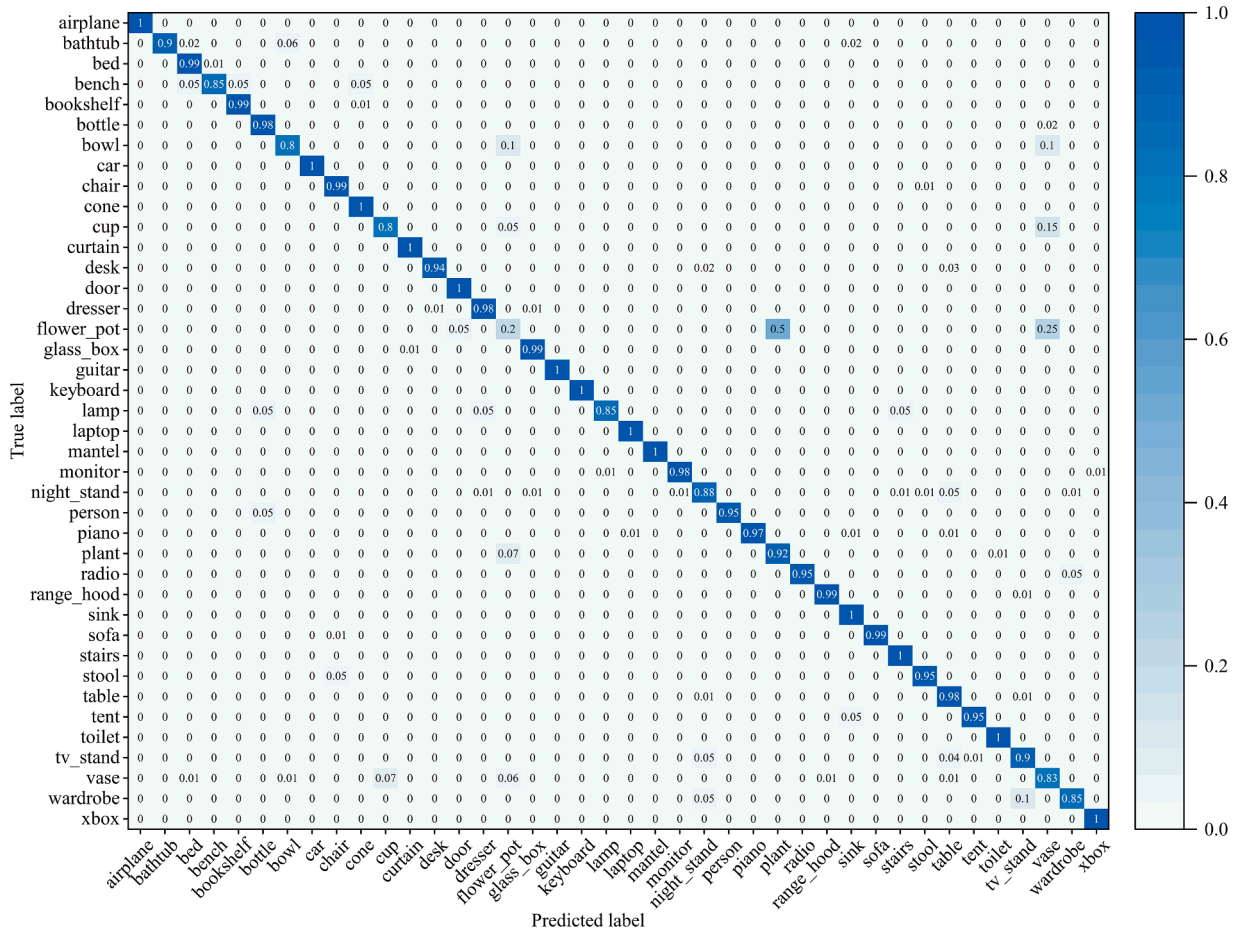


Fig. 6. Confusion matrix based on the proposed network.



Fig. 7. Examples of flower_pot and plant in ModelNet40.

fusion architecture, the network can already integrate cues from different scales through training, and explicitly learning scale weights does not improve performance. Therefore, we retain equal scale weighting as a simpler yet effective scale-level fusion strategy in our framework.

4.6. Overall component effectiveness analysis

Following the verification of the interaction and fusion module designs in [Sections 4.4](#) and [4.5](#), a stepwise ablation study is performed to evaluate the individual contribution of each component to the overall

performance. Swin_tiny is adopted as the baseline backbone, and the results are illustrated in Fig. 8.

From Fig. 8, the network shows a consistent stepwise improvement as each component is introduced. The baseline achieves 92.97%, and adding FPN improves it to 93.44%. This improvement suggests that multi-scale features enhance view representation. The introduction of either cross-group multi-head attention or Bi-FiLM further brings measurable gains. These results demonstrate the effectiveness of cross-group dependency modeling and conditional calibration in multi-view fusion. Combining FPN with cross-group attention or Bi-FiLM yields 94.21% (FPN + Cross-Attn) and 94.29% (FPN + Bi-FiLM), respectively.

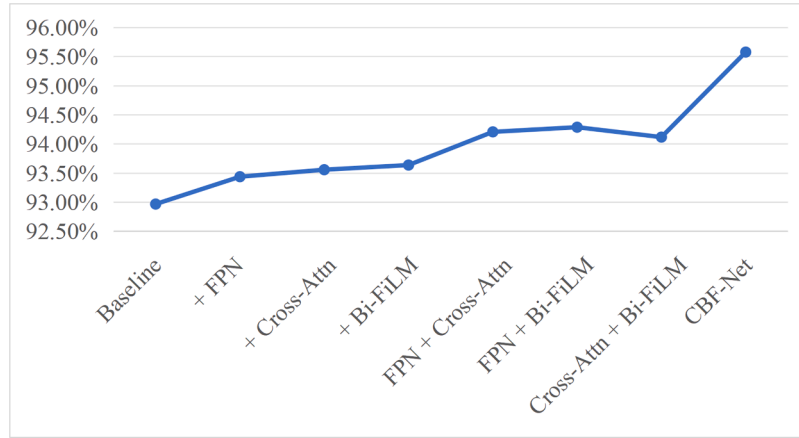


Fig. 8. Ablation study of overall component effectiveness.

Meanwhile, the interaction-modulation pair (Cross-Attn + Bi-FiLM) achieves 94.12%, which indicates complementary effects among the components. The complete model achieves 95.58%, representing a 2.61% improvement over the baseline. This indicates that rather than being simply stacked, these customized components work synergistically to address the core challenges of multi-view feature alignment and cross-view context calibration within our unified framework.

4.7. Ablation experiments with data augmentation strategies

The group of experiments includes Experiments 8–12, which compare the effects of different data augmentation strategies on the network proposed in this paper to process the task of multi-view classification of 3D models. In Experiment 8, Mixup data augmentation is applied simultaneously to two groups of views in each batch. In Experiment 9, CutMix data augmentation is applied simultaneously to two groups of views in each batch. Experiments 10–12 use dynamic selection of enhancement strategies. In Experiment 10, Mixup is selected with a probability of 30% and CutMix is selected with 70% probability for each batch. In Experiment 11, either Mixup or CutMix is equally selected with a probability of 50%. In Experiment 12, Mixup is chosen with a probability of 70% and CutMix with 30% for each batch. In this group of experiments, the classification performance is evaluated using the network model proposed in this paper. Training set is used to optimize the proposed network and the test set is used to verify it. To ensure experimental rigor and result stability, Experiments 8–12 are repeated independently five times using different random seeds. The box plot in Fig. 9 is plotted based on the final classification accuracy obtained from these five independent runs.

From Fig. 9, it can be seen that Experiment 11 has the highest median boxplot and the box is located at the top of the plot, showing its optimal performance. Experiments 8 and 9 have lower median accuracies and Experiment 9 has low outliers. This shows that the dynamic hybrid enhancement strategy is better than the single enhancement strategy. The boxplots of Experiments 10 and 12 are centered. Although their performance is superior to that of single strategies, they perform poorly compared to Experiment 11. This shows the effectiveness of the dynamic mixing strategy.

4.8. Additional dataset evaluation

To evaluate the performance of CBF-Net on additional datasets, experiments are conducted on ModelNet10 and ShapeNet. The official train/test split of ModelNet10 is used, including 3991 training models and 908 testing models from 10 object categories. For ShapeNet, a subset consisting of approximately 50,000 3D CAD models from 13 major categories is used. The category definitions and train/test split protocol

Table 8

Additional dataset evaluation results.

Dataset	Views	Accuracy (%)
ShapeNet	12	95.82
ModelNet10	12	95.59
ModelNet40	12	95.58

follow Choy et al. (2016), where 4/5 of the models are used for training and the remaining 1/5 are used for testing. For these additional experiments, each 3D model is represented by 12 rendered views, consistent with the ModelNet40 setting. The first six views are assigned to Group 1, and the remaining six views are assigned to Group 2. The same preprocessing procedure, view grouping strategy, image resolution, and CBF-Net architecture are used across datasets, except that the output dimension of the final classifier is adjusted according to the number of categories. No dataset-specific hyperparameter tuning is conducted for ModelNet10 or ShapeNet. The experimental results are shown in Table 8.

As shown in Table 8, CBF-Net achieves classification accuracies of 95.82%, 95.59%, and 95.58% on ShapeNet, ModelNet10, and ModelNet40, respectively. These results provide auxiliary evidence that the proposed framework can be applied to related rendered 3D classification datasets under the adopted 12-view input setting. The accuracy on ModelNet10 is close to that on ModelNet40, although ModelNet10 is generally considered easier because it contains fewer object categories. This observation should be interpreted with caution. In our experiments, ModelNet10 and ModelNet40 use the same CBF-Net architecture, 12-view rendering setting, preprocessing procedure, view grouping strategy, and training configuration, except that the final classifier is adjusted according to the number of categories. Therefore, the ModelNet10 result may be related to the unchanged experimental configuration, the adopted rendering protocol, and dataset-specific characteristics. The similar accuracy does not imply that ModelNet10 and ModelNet40 have comparable difficulty. Overall, the results on ModelNet10 and ShapeNet are presented as auxiliary evaluations on related rendered datasets, rather than as conclusive evidence of broad generalization.

4.9. Missing-view evaluation

To further evaluate the behavior of CBF-Net under incomplete-view conditions, a missing-view test is conducted on ModelNet40. During testing, the original 12-view input structure and fixed two-group partition are kept unchanged. Specifically, the first six views are assigned to Group 1, and the remaining six views are assigned to Group 2. A fixed proportion of the 12 rendered views is randomly masked before being

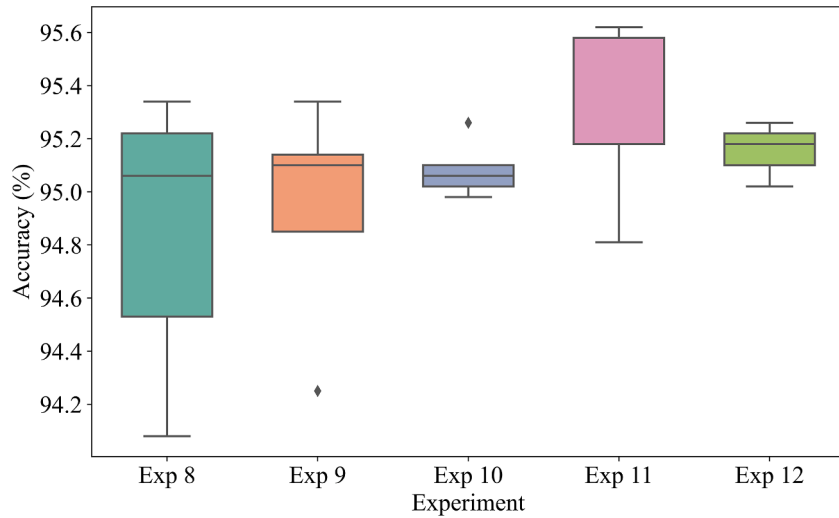


Fig. 9. Box plot of experimental results from five independent runs.

Table 9

Missing-view evaluation on ModelNet40.

Testing setting	Missing views	Accuracy (%)
Clean 12 views	0 / 12	95.58
25% views missing	3 / 12	95.21
50% views missing	6 / 12	93.85

fed into the network, while the model parameters and training protocol remain unchanged. The missing views are replaced by zero tensors after normalization. The reported accuracy is the average over three runs with different random seeds. The experimental results are shown in Table 9.

As shown in Table 9, CBF-Net achieves 95.21% accuracy when 25% of the views are randomly masked, with a decrease of 0.37 percentage points compared with the clean 12-view setting. When 50% of the views are masked, the accuracy decreases to 93.85%. These results provide an additional evaluation of CBF-Net under incomplete-view conditions and suggest that the proposed cross-group fusion framework can maintain reasonable performance when partial views are missing under the adopted testing protocol.

5. Conclusion

In this paper, we propose CBF-Net to address the challenges of insufficient inter-view interaction and inefficient feature fusion in multi-view 3D model classification. By integrating a multi-scale feature extraction architecture, a cross-group attention module, and a Bi-FiLM strategy, the proposed method improves the modeling of fine-grained geometric semantics and enhances multi-view feature fusion through bidirectional interactions between feature groups. Experimental results show that CBF-Net achieves competitive performance under the rendering protocol adopted in this paper, demonstrating the effectiveness of the proposed cross-group bidirectional fusion mechanism. Additional evaluations on ModelNet10, ShapeNet, and missing-view settings offer preliminary insights into the applicability of the proposed framework under related rendered datasets and incomplete-view conditions. Future work will focus on combining the proposed fusion mechanism with dynamic viewpoint selection strategies and further investigating the framework under more diverse evaluation settings.

CRedit authorship contribution statement

Xueyao Gao: Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation, Conceptualization; **Yali Shao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation, Conceptualization; **Chunxiang Zhang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation, Conceptualization; **Yongzeng Xue:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

ModelNet10 and ModelNet40 datasets used in this study can be downloaded from publicly available sources <http://modelnet.cs.princeton.edu/>.

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